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SUMMARY

MS Applied AI student (Stevens, 05/2027) targeting AI/GenAI Engineer roles. Built evaluated RAG and guarded text-to-SQL systems (hybrid retrieval, reranking, SQL guardrails, measured ablations) and medical-imaging deep learning pipelines. Technical writer on GenAI (Medium, Towards AI).

SKILLS

Languages: Python, SQL, R, C++
GenAI: RAG, Embeddings, Vector Search (Chroma), Hybrid Search (BM25 + Dense, RRF), Reranking, Prompt Engineering, OpenAI API, Guardrails
ML/DL: PyTorch, TensorFlow, Scikit-Learn, OpenCV, NumPy, Pandas, Apache Spark (PySpark)
Backend & Data: FastAPI, SQLAlchemy, REST APIs, PostgreSQL, Elasticsearch
Cloud & Tools: AWS (Bedrock, SageMaker, Comprehend, Transcribe, Polly, Textract, Kendra, S3, EC2, IAM), Docker, Git, Streamlit, Gradio
Concepts: LLM Evaluation (ablations, LLM-as-judge), AI Security Basics

PROJECTS

BankLens: Evidence-Grounded RAG over Bank Annual Reports | *Python, OpenAI (gpt-4o-mini, embeddings), Chroma, BM25, FastAPI, Streamlit, Docker* GitHub

- Built a RAG app for cited Q&A and claim verification (SUPPORTED / REFUTED / NOT_ENOUGH_INFO) over FY2024-25 annual reports of 4 major US banks; abstains when evidence is missing and highlights the source page.
- Ran a 6-configuration ablation (PDF parsing, table-aware chunking, hybrid dense + BM25 with RRF, bank/year metadata filter, cross-encoder reranking) on a 70-item golden set (40 questions, 30 claims).
- Improved Hit@5 from 0.567 to 0.700, MRR from 0.358 to 0.456 and answer correctness from 67.5% to 82.5%; metadata filtering was the largest low-cost gain.
- Validated the LLM judge on 25 hand-labelled answers (92% agreement on correctness, kappa 0.70) and found a lenient faithfulness judge (kappa 0.00), so faithfulness was reported as an upper bound.
- Added SQLite caching and cost caps (full evaluation under $1); containerized FastAPI and Streamlit with Docker Compose.

sqlsentry: Guarded Text-to-SQL Assistant | *Python, OpenAI (gpt-4o-mini), FastAPI, PostgreSQL, sqlglot, Streamlit, Docker* GitHub

- Built a natural-language analytics assistant over the Olist e-commerce dataset (~100k orders, 8 tables); the LLM only writes SQL and PostgreSQL computes every number, keeping answers exact and auditable.
- Designed layered guardrails: AST-based SQL guard (sqlglot) allowing only single SELECTs, a read-only DB role with 5s timeout and row cap, an append-only audit log, signed-token auth, login lockout and daily quotas.
- Implemented a self-repair loop (up to 2 retries on DB errors) and structured refusals; encoded business definitions in a schema card shared by the prompt and table allowlist, fixing a fan-out join that inflated revenue ~4.5%.
- Achieved 90-92% execution accuracy on 40 answerable questions (80-90% on a 10-question holdout) and 100% correct refusals on 8 unanswerable ones; 97 tests, including 70 adversarial guard tests, run without API calls.

Fair-Fracture: Bone Fracture Detection | *PyTorch, U-Net, YOLOv8, OpenCV, Gradio*

- Built a dual-model pipeline running U-Net segmentation (from scratch, joint BCE + Dice loss for <5% foreground imbalance; 99.26% pixel accuracy, 0.30 Dice) and YOLOv8-Nano detection of 17 fracture types (mAP50 0.364 overall, 0.891 on closed-simple) from a single X-ray.
- Trained on FracAtlas (4,083 radiographs) plus 2,000 Roboflow images; fused both outputs with confidence thresholding and OpenCV contour filtering in a side-by-side Gradio dashboard.

Skin Cancer Detection | *PyTorch, ResNet50, HAM10000, Gradio*

- Fine-tuned ImageNet-pretrained ResNet50 for benign vs. malignant classification on 10,015 images, using lesion-ID stratified splits to prevent patient-level leakage and weighted cross-entropy for a 60% class imbalance.
- Reached 82.00% malignant recall, 0.8813 AUC and 75.74% validation accuracy; deployed a Gradio app returning real-time diagnosis probabilities.

EXPERIENCE

Software Development Intern | *IDBI INTECH (subsidiary of IDBI Bank)* Mumbai, India | Dec 2023 - Jan 2024

- Contributed to a Hyperledger Fabric private blockchain for insurance claim processing across hospitals, insurers and government entities; tested transaction flows across peers, orderers, channels and chaincode.
- Engineered Logstash pipelines indexing application logs into Elasticsearch, and built Kibana dashboards for claim volumes and approval trends, transaction throughput, block times and system health.

Technical Writer | *Medium* Feb 2026 - Ongoing

- Published 2 LangChain articles in Towards AI (fundamentals; open-source models, prompt engineering).

EDUCATION

Stevens Institute of Technology, Hoboken, NJ | M.S. in Applied Artificial Intelligence Anticipated 05/2027
GPA: 3.5/4.0 | **Coursework:** Deep Learning, Machine Learning, Big Data Analytics, Probability & Stochastic Processes
University of Mumbai (SIES GST), India | B.E. in AI and Machine Learning Graduated 05/2025
GPA: 8.38/10.0 | **Coursework:** Data Structures & Algorithms, Computer Vision, NLP, Statistics